

IS5113 - GROUP PROJECT

The group projects are designed to engage students in critical thinking, ethical analysis, and practical application of AI in business contexts. These projects are suitable for groups of up to 8 students and encourage interdisciplinary collaboration, research, and presentation.

The final group project reports are due 1 week after the class presentation. The group project accounts for 50% of the final marks in this course. Students need to pass both the project and the final written examination to pass the course.

Each project contains 2 parts. Part 1 contains a research investigation on a current AI ethics topic. Each group will be assigned to work on one of the topics listed in Appendix A. This part of the report should contain no more than 15 pages.

Assessment Criteria (100 points)

1. Research Depth and Relevance (25 points)

- Quality and breadth of sources (academic, industry, legal, etc.)
- Relevance to AI ethics in business contexts
- Demonstrated understanding of key ethical theories and frameworks

2. Ethical Analysis and Argumentation (25 points)

- Identification of ethical issues and dilemmas
- Application of ethical principles (e.g., fairness, accountability, transparency)
- Quality of critical thinking and ethical reasoning

3. Practical Business Implications (20 points)

- Connection to real-world business practices
- Consideration of stakeholders (e.g., consumers, employees, regulators)
- Feasibility and impact of proposed ethical solutions or frameworks

4. Team Collaboration and Project Management (10 points)

- Evidence of equitable contribution from all team members
- Clear division of roles and responsibilities
- Use of project planning tools or methods (e.g., Gantt chart, meeting logs)

5. Presentation and Communication (20 points)

- Clarity and professionalism of the final report or presentation
- Use of visuals, data, and examples to support arguments
- Ability to engage the audience and respond to questions

Optional Bonus (5 points)

- Originality or innovation in approach
- Integration of interdisciplinary perspectives (e.g., law, psychology, design)
- Use of AI tools ethically in the research process

Part 2 contains policy recommendations tailored to a specific industry or sector. Each group will be assigned to work on one of the specific industries or sectors listed in Appendix B. Students should aim to explore ethical challenges within the chosen industry or sector and craft actionable policy recommendations that companies could adopt. Students should also compare their actionable policy recommendations with a specific company in the chosen industry or sector for benchmarking; this company may be chosen by the students freely. This part of the report should contain no more than 15 pages.

Assessment Criteria (100 points)

1. Ethical Policy Framework (30 points)

- Clarity and structure of the proposed policy
- Integration of ethical principles (e.g., fairness, accountability, transparency, privacy)
- Alignment with relevant legal, regulatory, and industry standards

2. Research and Evidence Base (15 points)

- Use of credible academic, legal, and industry sources
- Depth of analysis on the AI technology and its ethical implications
- Relevance and accuracy of supporting data or case studies

3. Stakeholder Analysis and Impact (15 points)

- Identification of key stakeholders (e.g., consumers, employees, regulators)
- Assessment of how the policy affects each stakeholder group
- Consideration of unintended consequences and mitigation strategies

4. Implementation Strategy (10 points)

- Feasibility of policy adoption within a business context
- Clear steps for implementation, monitoring, and enforcement
- Inclusion of mechanisms for accountability and revision

5. Team Collaboration and Project Management (10 points)

- Evidence of equitable contribution from all team members
- Clear division of roles and responsibilities
- Use of project planning tools or methods (e.g., Gantt chart, meeting logs)

6. Communication and Presentation (20 points)

- Professionalism and clarity of written report and/or presentation
- Use of visuals, executive summaries, and policy briefs
- Ability to defend and explain policy choices during Q&A

Optional Bonus (Up to 5 points)

- Originality or innovation in policy design
- Integration of interdisciplinary perspectives (e.g., law, psychology, design)
- Use of ethical AI tools in the policy development process

In the Group Project Report, the following information must be provided:

- The specific role and contribution of each team member
- Whether AI was used in preparing the report, and if so, how it was used, what tools were involved, and what was your contribution over and above those provided by AI

APPENDIX A: Current AI ethics topics

1. Algorithmic Bias in Hiring Platforms

Investigate how AI-driven recruitment tools may perpetuate bias. Analyze real-world cases and propose ethical frameworks for fair hiring.

2. Ethical Implications of AI in Financial Decision-Making

Explore how AI is used in credit scoring, loan approvals, and investment advice. Assess risks of discrimination and transparency.

3. AI and Consumer Privacy in E-Commerce

Evaluate how recommendation engines and targeted advertising use personal data. Propose ethical guidelines for data handling.

4. Surveillance and Workplace Monitoring

Analyze the ethical boundaries of AI-powered employee monitoring systems. Include legal and psychological perspectives.

5. Autonomous Vehicles and Corporate Responsibility

Study ethical dilemmas in self-driving car decision-making. Discuss liability, safety, and business accountability.

6. Deepfakes and Brand Reputation

Investigate how deepfake technology can impact businesses. Develop a policy framework for detection and response.

7. AI in Healthcare Startups: Ethics of Diagnosis

Examine ethical concerns in AI diagnostic tools used by health tech startups. Focus on accuracy, consent, and liability.

8. Ethical Supply Chain Management with AI

Explore how AI is used to optimize supply chains. Assess ethical risks related to labor, sustainability, and transparency.

9. AI and Misinformation in Marketing

Analyze how generative AI can be used (or misused) in marketing campaigns. Propose ethical standards for content creation.

10. Predictive Policing and Corporate Vendors

Study companies that sell predictive policing tools. Evaluate ethical concerns around racial profiling and accountability.

11. AI Ethics in Customer Service Chatbots

Assess how chatbots handle sensitive customer data and emotional interactions. Propose ethical design principles.

12. Corporate Governance and AI Ethics Boards

Research how companies are setting up internal AI ethics boards. Evaluate effectiveness and propose improvements.

13. AI in Gig Economy Platforms

Explore ethical issues in algorithmic management of gig workers (e.g., Uber, Deliveroo). Focus on fairness and autonomy.

14. Environmental Impact of AI Infrastructure

Investigate the carbon footprint of AI systems in business. Propose ethical sustainability strategies.

15. Cross-Cultural Ethics in Global AI Deployment

Analyze how ethical norms vary across cultures in AI applications. Develop a framework for multinational companies.

16. Ethics of Emotion Recognition in Retail

Study the use of AI to analyze customer emotions in stores. Discuss consent, manipulation, and psychological impact.

17. AI Ethics in Financial Trading Algorithms

Explore ethical risks in high-frequency trading and algorithmic decision-making. Focus on transparency and fairness.

18. Corporate Use of Generative AI for Internal Communications

Assess ethical risks of using generative AI for memos, reports, and HR communications. Discuss misinformation and trust.

APPENDIX B: AI ethics policy development for a specific industry or sector

1. Finance: Ethical AI Policy for Credit Scoring

Develop a policy to ensure fairness, transparency, and accountability in AI-driven credit scoring systems used by banks and fintech firms.

2. Healthcare: AI Ethics Policy for Diagnostic Tools

Create a policy framework for startups or hospitals using AI for medical diagnosis, focusing on consent, data privacy, and bias mitigation.

3. Retail: Responsible AI Use in Customer Behavior Prediction

Draft a policy for retailers that use AI to predict customer behavior, addressing ethical concerns about surveillance, manipulation, and data use.

4. Human Resources: AI Ethics in Automated Hiring

Design a policy for companies using AI in recruitment to prevent discrimination, ensure transparency, and uphold candidate rights.

5. Transportation: Ethics Policy for Autonomous Vehicle Decision-Making

Propose a policy for manufacturers or fleet operators to guide ethical decision-making in autonomous vehicle algorithms.

6. Education: AI Ethics in Adaptive Learning Platforms

Create a policy for ed-tech companies using AI to personalize learning, focusing on equity, data protection, and algorithmic transparency.

7. Legal Services: AI Ethics in Predictive Legal Analytics

Draft a policy for law firms or legal tech startups that use AI to predict case outcomes, ensuring fairness and preventing misuse.

8. Insurance: AI Ethics in Risk Assessment and Claims Processing

Develop a policy for insurers using AI to assess risk and process claims, addressing bias, explainability, and customer rights.

9. Real Estate: AI Ethics in Property Valuation and Market Forecasting

Create a policy for real estate platforms that use AI to value properties and forecast trends, with a focus on transparency and fairness.

10. Media & Entertainment: AI Ethics in Content Recommendation

Design a policy for AI-driven streaming platforms to recommend content, addressing filter bubbles, manipulation, and cultural bias.

11. Cybersecurity: AI Ethics in Threat Detection Systems

Draft a policy for cybersecurity firms using AI to detect threats, focusing on false positives, privacy, and human oversight.

12. Manufacturing: AI Ethics in Predictive Maintenance

Create a policy for manufacturers using AI to predict equipment failures, addressing worker safety, data integrity, and automation ethics.

13. Hospitality: AI Ethics in Guest Experience Personalization

Develop a policy for hotels using AI to personalize guest experiences, ensuring consent, privacy, and cultural sensitivity.

14. Logistics: AI Ethics in Route Optimization and Delivery Automation

Draft a policy for logistics firms that use AI to optimize routes and automate deliveries, addressing labor impacts and algorithmic fairness.

15. Gaming: AI Ethics in Player Behavior Prediction

Create a policy for gaming companies using AI to predict and influence player behavior, focusing on addiction, fairness, and consent.

16. Government Services: AI Ethics in Public Service Automation

Design a policy for government agencies using AI to automate services, ensuring equity, transparency, and public accountability.

17. Fashion: AI Ethics in Trend Forecasting and Design Automation

Propose a policy for fashion brands that use AI to forecast trends and automate design, addressing issues of originality, cultural appropriation, and sustainability.

18. Telecommunications: AI Ethics in Network Optimization and Customer Support

Develop a policy for telecom companies that use AI to manage networks and automate customer service, with a focus on fairness, privacy, and accessibility.